

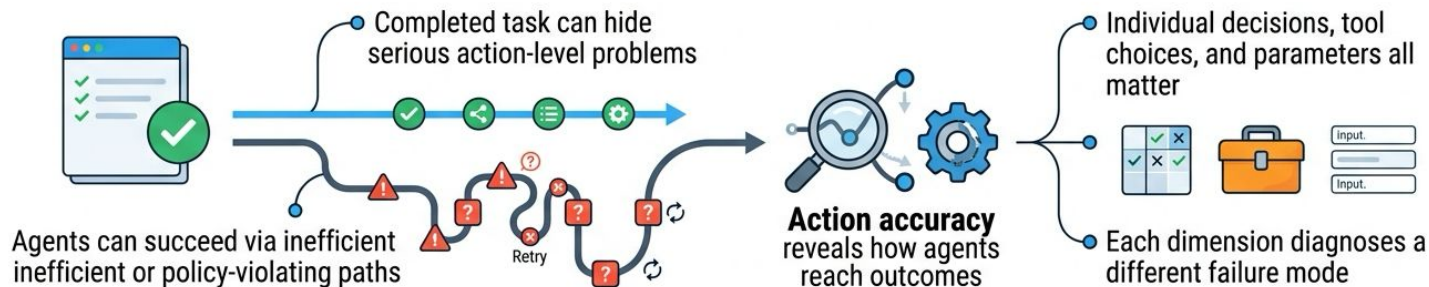


U.S. General Services
Administration

Chapter 3.8: Action Accuracy Metrics

US General Service Administration
AI Community of Practice - March 2026

Dimensions of Action Accuracy



[Four Dimensions of Action Accuracy]

Tool selection accuracy



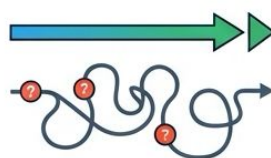
did the agent pick the right tool?

Parameter accuracy



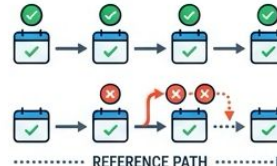
are argument values correct and grounded?

Execution path validity



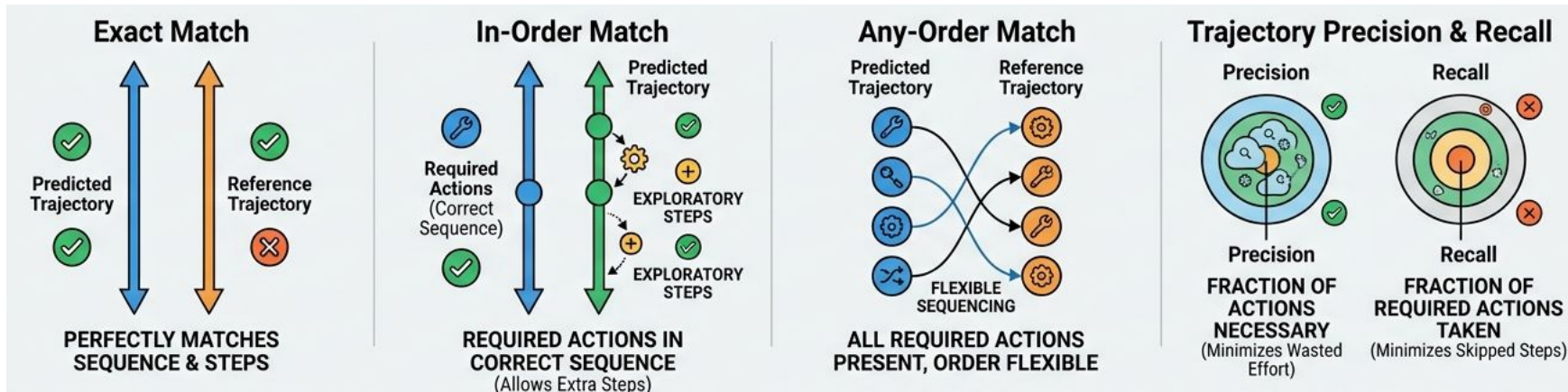
was the route efficient and logical?

Action trajectory quality



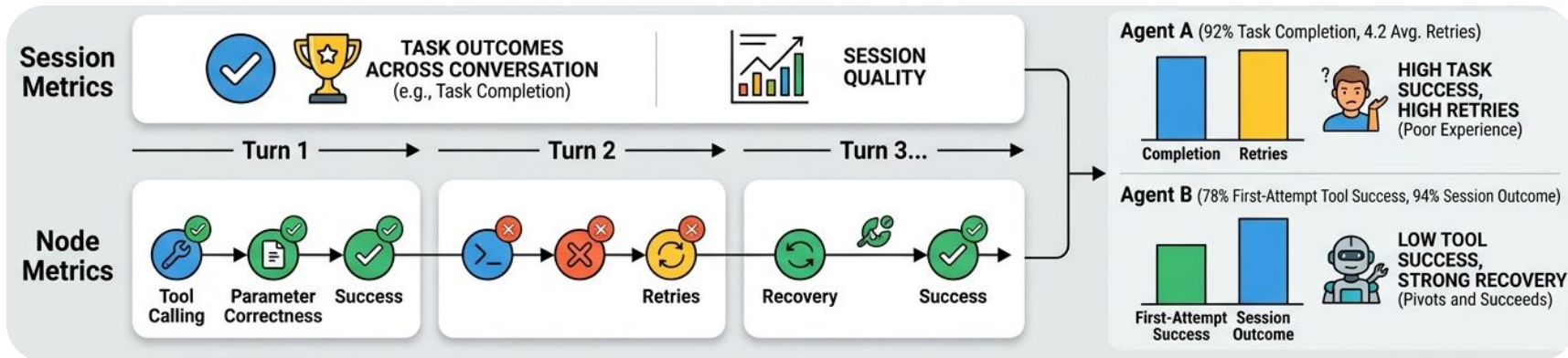
does the full sequence match expectations?

Trajectory Evaluation Metrics



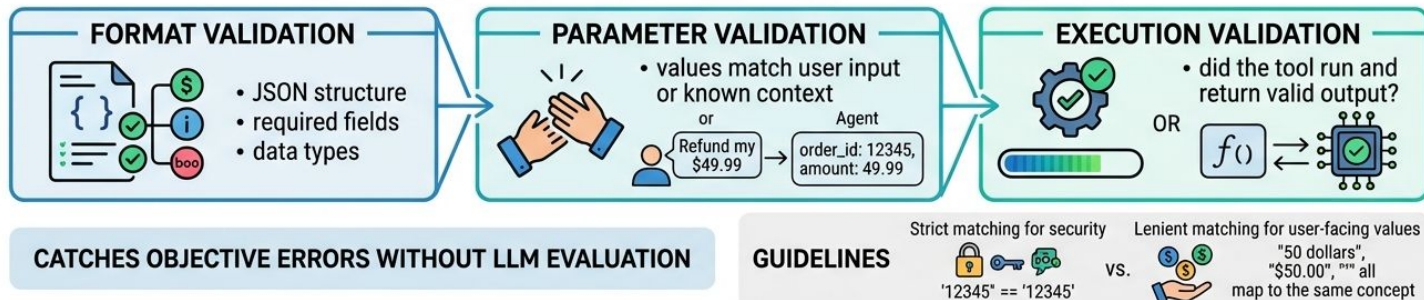
- Exact match: predicted trajectory perfectly matches reference
- In-order match: all required actions present in correct sequence
- Any-order match: all required actions, flexible sequencing
- Trajectory precision: fraction of actions that were necessary
- Trajectory recall: fraction of required actions actually taken

Multi-Turn Consistency and Layered Evaluation

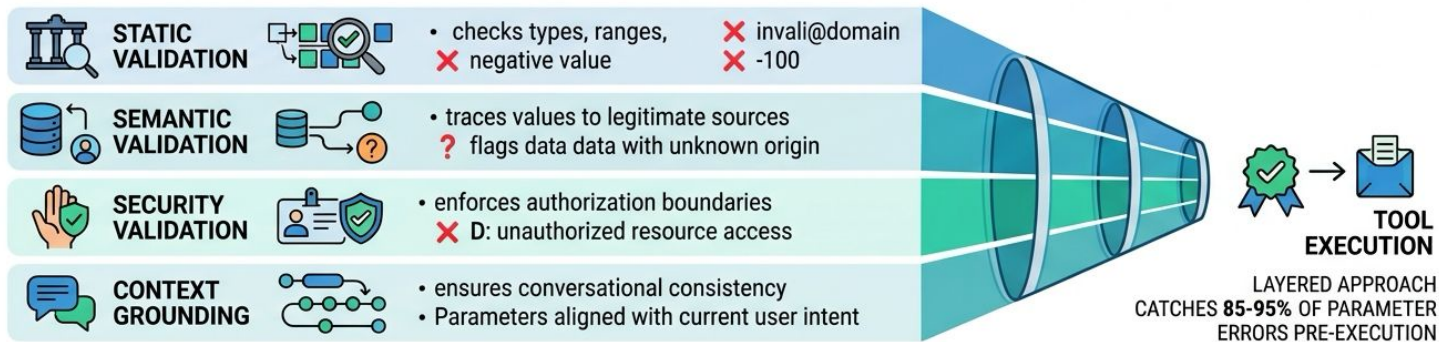


- Single-action metrics miss degradation across conversation turns
- Session metrics: task outcomes across entire conversations
- Node metrics: individual action correctness at each step
- An agent may complete 92% of tasks but average 4.2 retries
- Layered view reveals both error frequency and recovery ability

Programmatic Validation



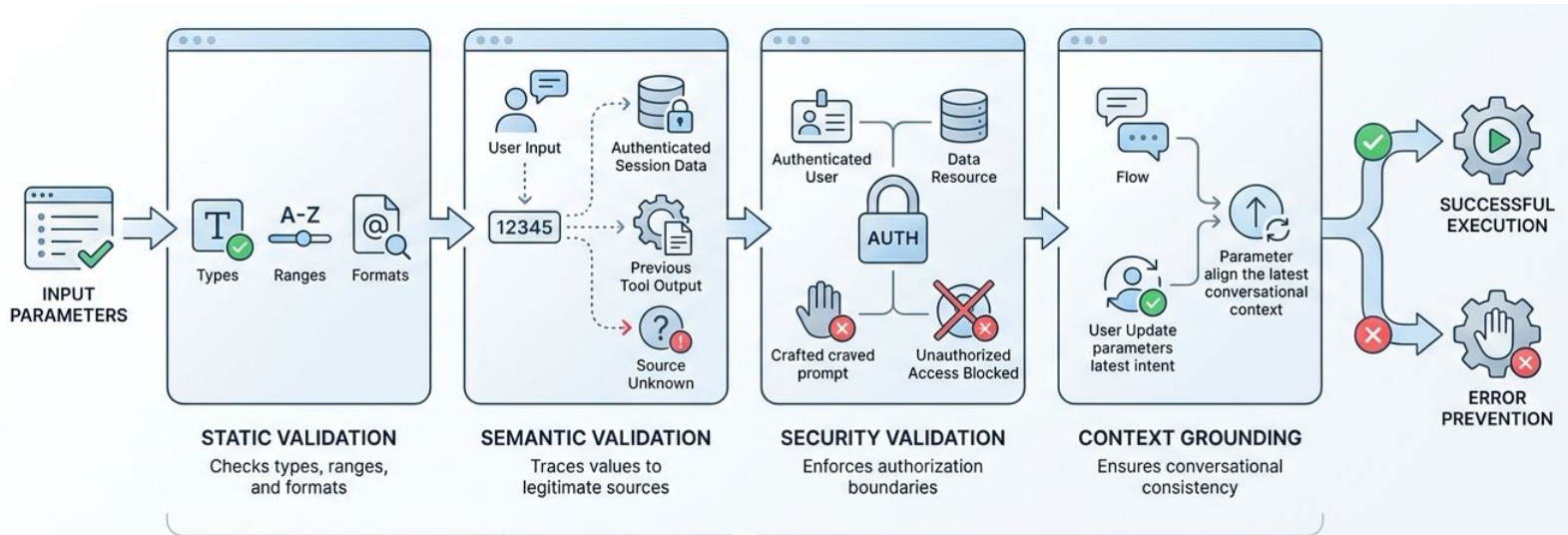
THE PARAMETER VALIDATION PIPELINE



LLM-as-Judge and BFCL Metrics

- LLM evaluators assess subjective action quality with rubrics
 - Parameter Hallucination Rate: agents invent nonexistent parameters
 - Parameter Missing Rate: required arguments omitted from calls
 - Progress Rate: how far agents get before the first error
 - Combined metrics pinpoint whether selection or usage is the problem
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The Parameter Validation Pipeline

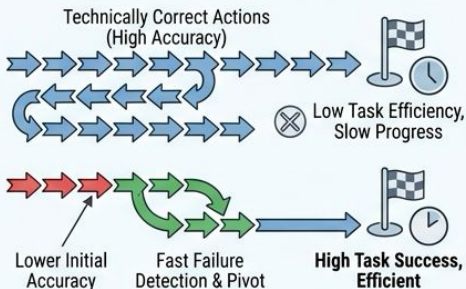


85-95%

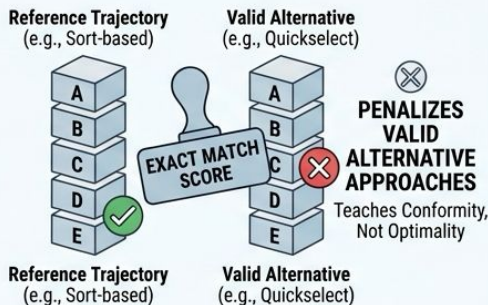
Layered approach catches 85-95% of parameter errors pre-execution.

Common Pitfalls -- Metrics That Mislead

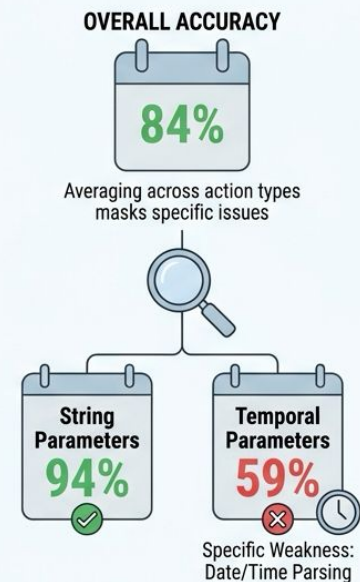
HIGH ACTION ACCURACY $\neq$ TASK SUCCESS



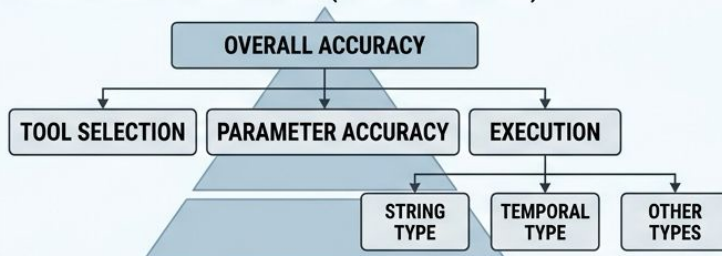
EXACT MATCH TRAP



AVERAGING HIDES WEAKNESSES

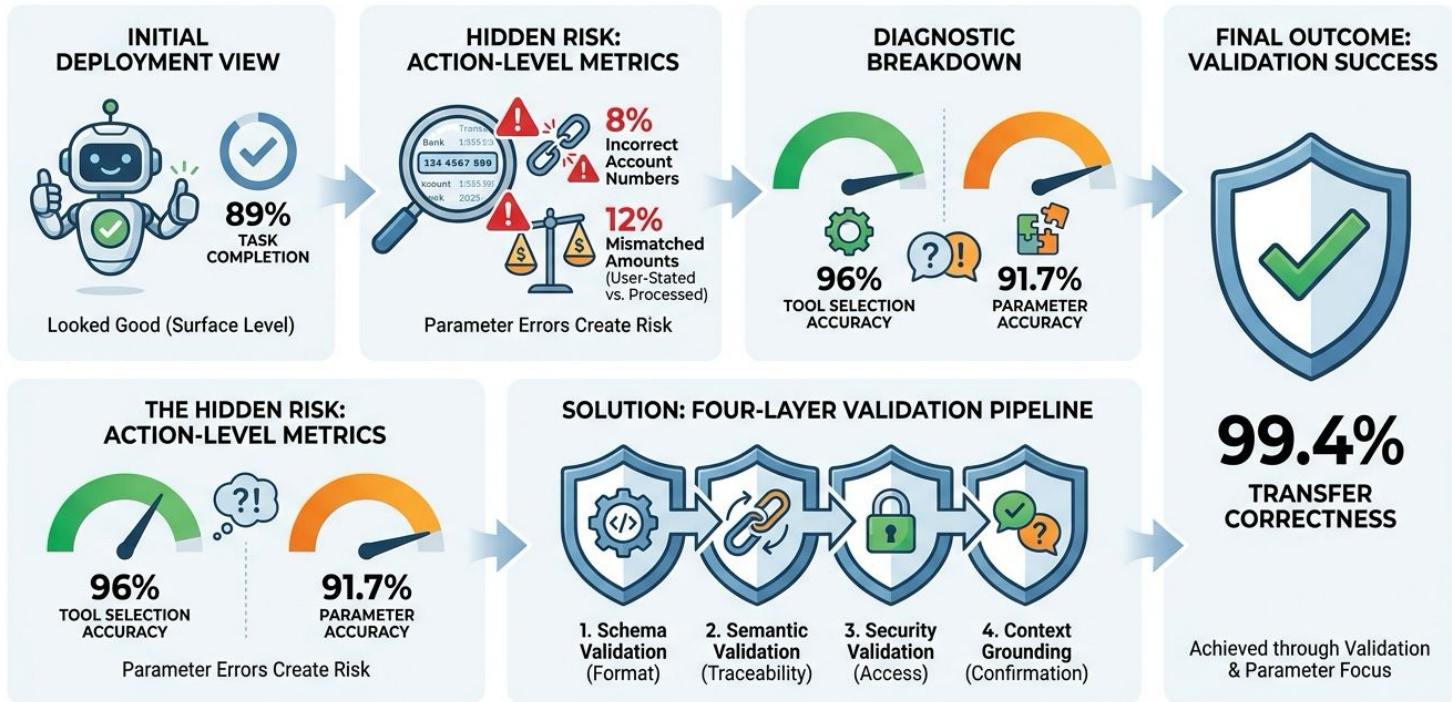


SOLUTION: HIERARCHICAL METRIC REPORTING (FOR DIAGNOSTICS)



Break down metrics to pinpoint bottlenecks (e.g., Temporal accuracy jumps from 59% to 72% after fix). Directly attribute improvements.

Example -- Parameter Accuracy as Risk Control



Key Takeaways and Bridge to Chapter 3.9

- Action accuracy reveals how agents succeed, not just whether they do
 - Four dimensions: tool selection, parameters, execution path, trajectory
 - Layer your validation: static, semantic, security, context grounding
 - Report metrics hierarchically -- averages hide actionable insights
 - Next: Chapter 3.9 evaluates reasoning quality behind the actions
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